AI Assissted Coding
Lab Assignment 10.1

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1. Syntax and Logic Errors

   Task:

   Use AI to identify and fix syntax and logic errors in a faulty Python script.

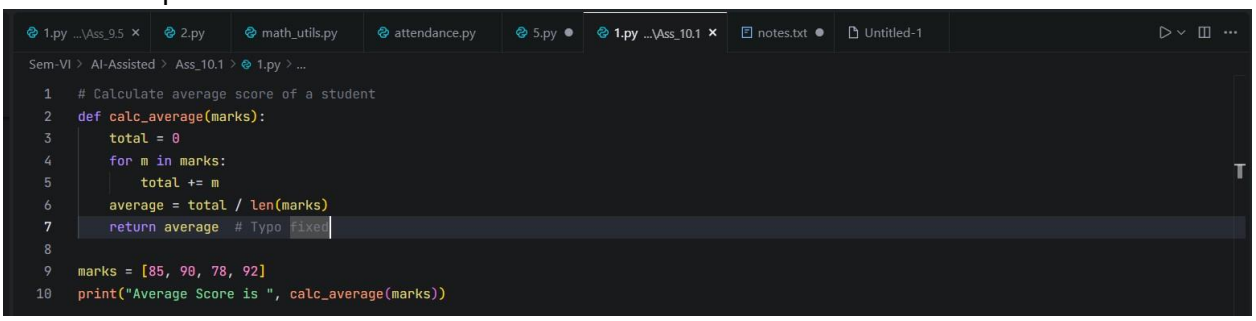
   Sample Input Code:

```
# Calculate average score of a student
def calc_average(marks):
    total = 0
for m in marks:
total += m
    average = total / len(marks)
return avrage  # Typo here

marks = [85, 90, 78, 92] print("Average Score
is ", calc_average(marks)
```

   Expected Output:

   - Corrected and runnable Python code with explanations of the fixes. Fixed Code:



2. PEP 8 Compliance

   Task:

   Use AI to refactor Python code to follow PEP 8 style guidelines.

   Sample Input Code:

```
def area_of_rect(L,B) : return L*B
print(area_of_rect(10,20))
```

Expected Output:

- Well-formatted PEP 8-compliant Python code.

Well Formatted Code:

```python
def area_of_rect(length: float, breadth: float) -> float:
    """
    Returns the area of a rectangle.

    Parameters:
    length (float): The length of the rectangle.
    breadth (float): The breadth of the rectangle.

    Returns:
    float: The area of the rectangle.
    """
    return length*breadth
print(area_of_rect(10,20))
```

3. Readability Enhancement Task:

Use AI to make code more readable without changing its logic.

Sample Input Code:

```python
def c(x,y):
    return x*y/100 a=200 b=15
print(c(a,b))
```

Expected Output:

- Python code with descriptive variable names,

inline comments, and clear formatting. Enhanced Readability:

```python
def calculate_percentage(x: float, y: float) -> float:
    """
    Returns the result of multiplying x and y, then dividing by 100.

    Parameters:
    x (float): The first number to be multiplied.
    y (float): The second number to be multiplied.

    Returns:
    float: The result of (x * y) / 100.
    """
    return x*y/100
a = 200
b = 15
print(calculate_percentage(a,b))
```

4. Refactoring for Maintainability Task:

Use AI to break repetitive or long code into reusable functions. Sample Input Code:

```python
students = ["Alice", "Bob", "Charlie"]
print("Welcome", students[0])
print("Welcome", students[1])
print("Welcome", students[2])
```

Expected Output:

- Modular code with reusable functions.

Modular Reusable Function:

```python
# Use AI to break repetitive or long code into reusable functions
def welcome_student(name: str) -> None:
    """
    Prints a welcome message for the given student name.

    Parameters:
    name (str): The name of the student.
    """
    print("Welcome", name)

students = ["Alice", "Bob", "Charlie"]
for student in students:
    welcome_student(student)
```

5. Performance Optimization Task:

Use AI to make the code run faster. Sample

Input Code:

```python
# Find squares of numbers nums = [i for
i in range(1,1000000)] squares = [] for
n in nums:
    squares.append(n**2)
print(len(squares))
```

Expected Output:

Optimized code using list comprehensions or vectorized operations.

Optimized Code:

```python
# using list comprehension
nums = [i for i in range(1,1000000)]
squares = [n**2 for n in nums]
print(len(squares))

# using vectorized operations
import numpy as np
nums = np.array(nums)
squares = nums**2
print(len(squares))
```

6. Complexity Reduction Task:

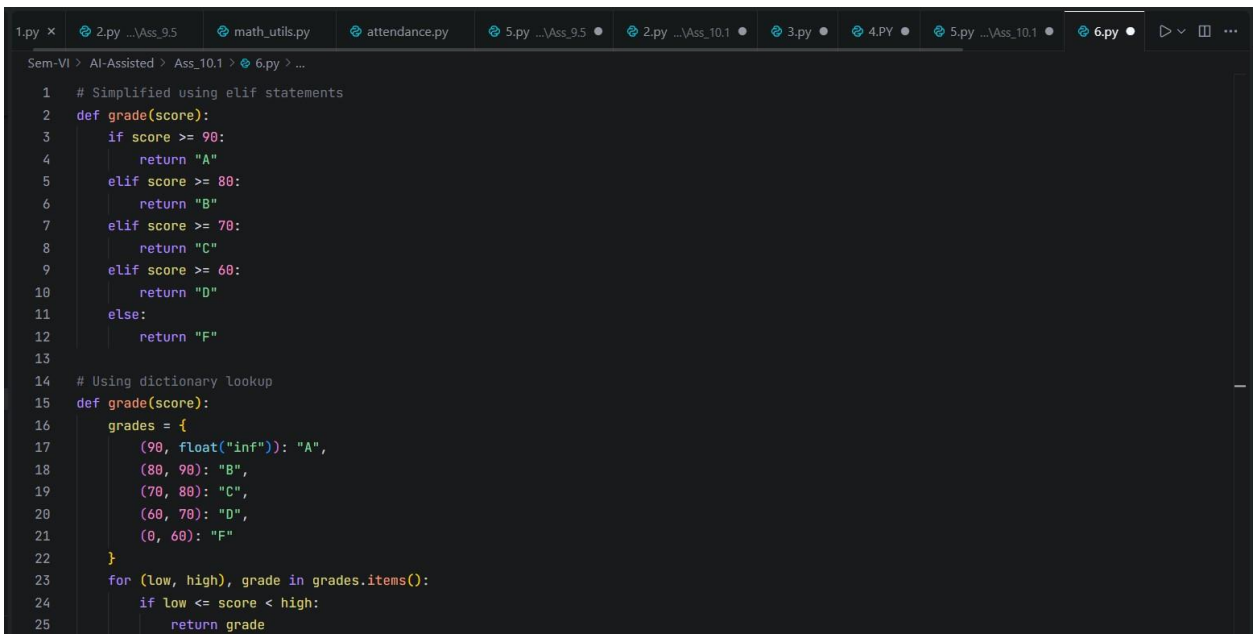
Use AI to simplify overly complex logic. Sample

Input Code:

```python
def grade(score):
if score >= 90:
return "A"     else:
```

```
        if score >= 80:
return "B"         else:
if score >= 70:
return "C"
else:
                if score >= 60:
return "D"                 else:
                    return "F"
```

Expected Output:

- Cleaner logic using elif or dictionary mapping. Simplified

code